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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Elective (DSE) - MG4DSETTM200 – AVIATION MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Objective Type Questions, Answer all questions Each question carries 1 marks

1.What is overbooking in airlines?[U] / [CO3]

Ans: Selling more seats than available

2.Airports generate revenue through landing fees, parking charges, and whichother source? [K] / [CO2]

Ans: Retail and concession services

3.Airlines must comply with rules for: [U] / [CO4]

Ans: Safety

4.Ethical issues in aviation relate to: [U] / [CO4]

Ans: Fair treatment of passengers

5.What is the primary role of an airline operations control center in daily flight operations? [K] / [CO1]

Ans: Monitoring and coordinating flights in real time

Part B

Short Answer QuestionsAnswer any5questionsEach question carries2marks

6.Analyse the role of aviation in international trade expansion. [An] / [CO1]

Ans: Aviation enables rapid transport of goods and people, boosting international trade and global supply chains. It supports economic growth by connecting markets and facilitating business activities worldwide.

7.Describe the functions of terminal control in busy airport airspace.[A] / [CO2]

Ans: Terminal control manages aircraft in the airport’s surrounding airspace to ensure safe separation. It coordinates arrivals and departures for smooth and efficient traffic flow.

8.Explain FAA certification process.[U] / [CO3]

Ans: The Federal Aviation Administration (FAA) certification process ensures aircraft, pilots, and operators meet safety standards. It includes testing, inspection, and approval before operations are allowed.

9.Define globalization in aviation context.[K] / [CO3]

Ans: Globalization in aviation refers to the worldwide connectivity of air transport networks. It enables movement of people and goods across countries, promoting trade and tourism.

10.How does runway management support safe airport operations?[A] / [CO2**]**

Ans: Runway management ensures safe takeoffs and landings by coordinating aircraft movements. It reduces risks of collisions and improves airport efficiency.

11.State the importance of fuel efficiency in fleet planning.[I] / [CO1]

Ans: Fuel efficiency reduces airline operating costs and increases profitability. It also helps in minimizing environmental impact through lower emissions.

12.What is corporate social responsibility (CSR) in aviation?[U] / [CO4]

Ans: Corporate Social Responsibility (CSR) in aviation refers to ethical practices towards society and the environment. It includes sustainability, safety, and community support initiatives.

13.Analyse the structure of ICAO governance.[An] / [CO3**]**

Ans: The International Civil Aviation Organization (ICAO) has a governance structure with the Assembly, Council, and Secretariat. These bodies ensure global aviation standards, policy-making, and implementation.

Part C

Short Essay Questions Answer any 5 questions Each question carries 5 marks

14.Discuss Urban Air Mobility (UAM) and its impact on city transportation systems. [U] / [CO4]

Ans: Urban Air Mobility (UAM) refers to the use of small, often electric aircraft for short-distance travel within cities. It can reduce traffic congestion and improve urban transportation efficiency.

15.Evaluate the impact of GPS-based navigation systems on flight safety andoperational efficiency.[E] / [CO4]

Ans: GPS-based navigation systems improve flight safety by providing accurate positioning and route guidance. They also enhance operational efficiency by optimizing flight paths and reducing fuel consumption.

16.Explain the role of airport security screening in preventing unlawful interferencein aviation.[A] / [CO2]

Ans: Airport security screening helps detect prohibited items and prevent unlawful interference like hijacking or terrorism. It ensures passenger safety and maintains secure aviation operations.

17.Explain hub, secondary, and feeder airport classifications with examples

[An] / [CO2]

Ans: Hub airports are major centers connecting many routes (e.g., large international airports), while secondary airports serve regional traffic. Feeder airports connect smaller areas to hubs, supporting overall network connectivity.

18.What are the main types of aviation emergencies?[U] / [CO3]

Ans: Main types of aviation emergencies include technical failures, medical emergencies, and security threats. Natural disasters and weather-related issues are also common aviation emergencies.

19.Explain how technological innovations support environmental sustainability inaviation[A] / [CO4]

Ans: Technological innovations like fuel-efficient aircraft and sustainable aviation fuels reduce environmental impact. Advanced systems also help lower emissions and improve energy efficiency.

20.Evaluate how SMS helps in identifying and mitigating operational risks inaviation. [E] / [CO3]

Ans: Safety Management Systems (SMS) help identify potential risks through continuous monitoring and reporting. They support mitigation strategies to improve overall aviation safety.

21.What is aviation economics?[K] / [CO3]

Ans: Aviation economics studies the financial and economic aspects of the aviation industry. It includes cost management, pricing, demand, and airline profitability.

Part D

Long Essay QuestionsAnswer any2questionsEach question carries15marks

22.Critically examine the role of passenger terminal facilities in improving airportefficiency and passenger satisfaction.[E] / [CO2]

Ans: Passenger terminal facilities improve airport efficiency by ensuring smooth passenger flow through check-in, security, and boarding processes. Modern amenities like lounges, signage, and digital services enhance passenger comfort and satisfaction. However, overcrowding and poor management can negatively affect both efficiency and traveler experience.

23.Explain the role of budgeting in aviation cost management.[A] / [CO3]

Ans: Budgeting in aviation helps control costs by planning and allocating financial resources effectively. It enables airlines and airports to monitor expenses and improve profitability. Proper budgeting also supports decision-making and long-term financial stability.

24.Discuss the contribution of technological advancements such as compositematerials, fuel-efficient engines, and electric aircraft to sustainable aviation.[U] / [CO4]

Ans: Technological advancements like composite materials reduce aircraft weight and improve fuel efficiency. Fuel-efficient engines and electric aircraft lower emissions and environmental impact. These innovations play a key role in achieving sustainable aviation development.

25.Critically examine the challenges faced by civil aviation in the modern aviation environment.[E] / [CO1]

Ans: Civil aviation faces challenges such as rising fuel costs, environmental concerns, and strict safety regulations. Increasing competition and technological changes also create operational pressures. Additionally, global crises like pandemics and geopolitical issues impact the stability of the aviation industry.